

Full-System Backstepping Landing Control

Mathematical Derivation and Stability Analysis

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Abstract

This document derives and formally analyses a **full-system backstepping** landing controller for a planar thrust-vector-controlled (TVC) rocket. A single composite Lyapunov function simultaneously covers position, attitude, and the nozzle actuator. The derivation proceeds in four backstepping steps, producing virtual controls for desired pitch angle, angular rate, and nozzle deflection, before arriving at the real nozzle command. Global practical stability under actuator saturation is proved by establishing that the closed-loop system is **input-to-state stable (ISS)** with respect to actuator clipping errors. ISS is a property we prove for our specific system — not an assumed result — by constructing a Lyapunov inequality of the form $\dot{V} \leq -c\|z\|^2 + D'$, where D' depends on the magnitude of the clipping errors. This guarantees the state remains bounded whenever the disturbances are bounded. The full proof with explicit quantitative bounds is in Section 10.5.

1 Plant Description

The system is a planar TVC rocket in the vertical plane. A single engine deflects by angle δ from the body axis, and δ is governed by a first-order actuator. The rocket operates in the low-speed aerodynamic regime, so aerodynamic drag, normal force, and a pitching moment act on it. All aerodynamic coefficients are assumed known.

State vector and control inputs

$$s = [x, y, \dot{x}, \dot{y}, \vartheta, \dot{\vartheta}, \delta]^T \in \mathbb{R}^7$$

Control inputs: $\sigma \in [\sigma_{\min}, 1]$ (throttle) and $\delta_{\text{cmd}} \in [-\delta_{\max, \text{cmd}}, \delta_{\max, \text{cmd}}]$ (nozzle command).

System parameters

Symbol	Meaning	Units
m	Total mass (constant)	kg
J	Moment of inertia about CoM (constant)	kg m ²
l_{cp}	Distance from CoM to nozzle exit	m
g	Gravitational acceleration, 9.81	m/s ²
$F_{\max}$	Maximum thrust	N
$\delta_{\max}$	Physical nozzle deflection limit	rad
τ_{δ}	Nozzle actuator time constant	s
ρ	Air density, 1.225	kg/m ³
S_m	Reference cross-sectional area	m ²
l	Reference length	m
C_x	Axial drag coefficient, 0.358	—
$C_{y\alpha}$	Normal force coefficient, 0.05403	deg ⁻¹
$C_{m\alpha}$	Pitching moment coefficient, 1.054	rad ⁻¹

2 Simplifying Assumptions

A1. Constant mass. $\dot{m} = 0$, $\dot{J} = 0$, $\dot{l}_{cp} = 0$.

In a real rocket, fuel is consumed during the burn, reducing mass and shifting the centre of mass. Here we assume the landing burn is short enough that mass change is negligible — a standard assumption for terminal descent from a few hundred metres altitude.

A2. Low-speed aerodynamic regime. $V \leq 100$ m/s (Mach < 0.3): $C_x = 0.358$, $C_{m\alpha} = 1.054$ rad $^{-1}$, $C_{y\alpha} = 0.05403$ deg $^{-1}$.

Below Mach 0.3 compressibility effects are small and aerodynamic coefficients are approximately constant. This allows us to use fixed coefficient values throughout the landing manoeuvre rather than scheduling them with Mach number.

A3. All parameters known. $C_{m\alpha}$, C_x , $C_{y\alpha}$, J , l_{cp} , τ_δ are all known.

A4. Small nozzle deflection for translational linearisation. $|\delta| \leq \delta_{\max} \leq 0.25$ rad, so:

$$\sin(\vartheta + \delta) \approx \sin \vartheta + \delta \cos \vartheta \quad (1)$$

$$\cos(\vartheta + \delta) \approx \cos \vartheta - \delta \sin \vartheta \quad (2)$$

Error: $|\sin(\vartheta + \delta) - (\sin \vartheta + \delta \cos \vartheta)| \leq \delta^2/2 < 3\%$ for $|\delta| \leq 0.25$ rad.

The nozzle angle is physically limited to a narrow range (here ± 0.25 rad $\approx \pm 14^\circ$) by the gimbal mechanism. Because δ is small, the first-order Taylor expansion of $\sin(\vartheta + \delta)$ and $\cos(\vartheta + \delta)$ around $\delta = 0$ is accurate to within 3%. This separates the dominant thrust direction (governed by body pitch ϑ) from the small torque contribution of the nozzle deflection δ , making the equations tractable for backstepping.

A5. Small pitch error for rotational linearisation. $|z_\vartheta| = |\vartheta - \vartheta^*| \ll 1$ rad during the controlled transient, so $\sin(\vartheta^* + z_\vartheta) \approx \sin \vartheta^* + \cos \vartheta^* \cdot z_\vartheta$. This allows the position dynamics to be written in terms of z_ϑ explicitly, enabling a single Lyapunov function.

The backstepping design requires writing the time derivative of V_{pos} as an explicit function of the pitch error z_ϑ . This is only possible if we can linearise $\sin \vartheta$ around the desired pitch ϑ^ . The assumption holds whenever the attitude controller is fast enough to keep the tracking error small — which is enforced by the inner-loop gains k_ϑ and k_ω . It is the only assumption that creates a conservatism gap: if $|z_\vartheta|$ ever grows large (e.g., at the very start with a 30° initial tilt), the linearisation error acts as an additional bounded disturbance absorbed into the ISS bound.*

A6. Minimum throttle. $\sigma \geq \sigma_{\min} > 0$, so $g_2 = Fl_{cp}/J > 0$ always.

The control authority of the nozzle enters the rotational dynamics as $g_2 = Fl_{cp}/J$. If the engine were shut off ($\sigma = 0$), $g_2 = 0$ and the virtual control α_2 in Step 3 would involve division by zero (see equation (VC2)). Enforcing a minimum throttle $\sigma_{\min}$ keeps g_2 bounded away from zero and the controller well-defined at all times.

A7. Full state measurement. $(x, y, \dot{x}, \dot{y}, \vartheta, \dot{\vartheta}, \delta)$ is measurable without noise.

3 Equations of Motion

3.1 Full Nonlinear Equations

The thrust vector makes angle $(\vartheta + \delta)$ from the vertical in the inertial frame. Projecting thrust and body-frame aerodynamic forces (X_b axial drag, Y_b normal force) onto the inertial axes, and including the nozzle actuator and pitch moment, the exact equations of motion are:

$$m\ddot{x} = F \sin(\vartheta + \delta) + X_b \sin \vartheta + Y_b \cos \vartheta \quad (\text{NL1})$$

$$m\ddot{y} = F \cos(\vartheta + \delta) - mg + X_b \cos \vartheta - Y_b \sin \vartheta \quad (\text{NL2})$$

$$J\ddot{\vartheta} = F l_{cp} \sin \delta + C_{m\alpha} \alpha q_\infty S_m l \quad (\text{NL3})$$

$$\tau_\delta \dot{\delta} = -\delta + \delta_{\text{cmd}} \quad (\text{NL4})$$

with $F = \sigma F_{\max}$, $X_b = -C_x q_{\infty} S_m$, $Y_b = C_{y\alpha} \alpha q_{\infty} S_m$, $q_{\infty} = \frac{1}{2} \rho V^2$, $V = \sqrt{\dot{x}^2 + \dot{y}^2}$, and angle of attack:

$$\alpha = \vartheta - \text{atan2}(\dot{x}, \dot{y})$$

Note that (NL1)–(NL2) are fully coupled: the translational accelerations depend on the nozzle angle δ through $\sin(\vartheta + \delta)$ and $\cos(\vartheta + \delta)$, and on pitch ϑ through the aerodynamic projections. The rotational equation (NL3) is driven by δ through the moment arm l_{cp} .

3.2 Small- δ Linearisation (Assumption A4)

Applying A4 to (NL1)–(NL2):

$$m\ddot{x} = F \sin \vartheta + F \delta \cos \vartheta + X_b \sin \vartheta + Y_b \cos \vartheta \quad (\text{T1})$$

$$m\ddot{y} = F \cos \vartheta - mg - F \delta \sin \vartheta + X_b \cos \vartheta - Y_b \sin \vartheta \quad (\text{T2})$$

with $F = \sigma F_{\max}$, $X_b = -C_x q_{\infty} S_m$, $Y_b = C_{y\alpha} \alpha q_{\infty} S_m$, $q_{\infty} = \frac{1}{2} \rho V^2$, and angle of attack $\alpha = \vartheta - \text{atan2}(\dot{x}, \dot{y})$.

This is the angle between the body axis and the velocity vector, measured positive nose-up. At zero airspeed ($V < V_{\min} \approx 0.1$ m/s), aerodynamic forces vanish and α is set to ϑ by convention.

Structural observation. In (T1)–(T2), ϑ drives position through the dominant terms $F \sin \vartheta$, $F \cos \vartheta$, while δ influences position through the smaller coupling terms $F \delta \cos \vartheta$, $-F \delta \sin \vartheta$. Retaining these coupling terms is what makes the full-backstepping approach stronger than a simple outer-inner loop design.

Applying A4 to the rotational equation ($\sin \delta \approx \delta$) and keeping the actuator equation exact:

$$J\ddot{\vartheta} = F l_{cp} \delta + C_{m\alpha} \alpha q_{\infty} S_m l \quad (\text{R1})$$

$$\tau_{\delta} \dot{\delta} = -\delta + \delta_{\text{cmd}} \quad (\text{A1})$$

Define the shorthand:

$$f_2 = \frac{C_{m\alpha} \alpha q_{\infty} S_m l}{J} \quad (3)$$

$$g_2 = \frac{F l_{cp}}{J} > 0 \quad (4)$$

so that $\ddot{\vartheta} = g_2 \delta + f_2$. Note that $C_{m\alpha} > 0$ means the rocket body is **aerodynamically unstable** — a positive angle of attack produces a nose-up pitching moment, which amplifies the perturbation. The controller must actively cancel f_2 via the (VC2) term $+\dot{\alpha}_1 - f_2$ in the nozzle reference.

4 Control Objective and Error Coordinates

Landing target:

$$s^* = (x_d, 0, 0, 0, 0, 0, 0)^T$$

Error coordinates:

$$e_x = x - x_d, \quad e_y = y \quad (5)$$

$$\dot{e}_x = \dot{x}, \quad \dot{e}_y = \dot{y} \quad (6)$$

The full objective: $(e_x, e_y, \dot{x}, \dot{y}, \vartheta, \dot{\vartheta}, \delta) \rightarrow 0$ as $t \rightarrow \infty$.

5 Unified Lyapunov Function — Structure and Motivation

The full-backstepping Lyapunov function is built up one term at a time — one $\frac{1}{2}z_i^2$ per step — until the entire state is covered:

$$V = \underbrace{\frac{1}{2}k_{px}e_x^2 + \frac{1}{2}\dot{x}^2 + \frac{1}{2}k_{py}e_y^2 + \frac{1}{2}\dot{y}^2}_{V_{\text{pos}}} + \underbrace{\frac{1}{2}z_{\vartheta}^2}_{\text{Step 2}} + \underbrace{\frac{1}{2}z_{\omega}^2}_{\text{Step 3}} + \underbrace{\frac{1}{2}z_{\delta}^2}_{\text{Step 4}} \quad (\text{V})$$

where the backstepping errors are defined sequentially:

$$z_{\vartheta} = \vartheta - \vartheta^*, \quad z_{\omega} = \dot{\vartheta} - \alpha_1, \quad z_{\delta} = \delta - \alpha_2$$

- z_{ϑ} : how far the actual pitch ϑ is from the desired pitch ϑ^* computed in Step 1. If $z_{\vartheta} = 0$, the rocket points exactly where the position controller wants it to.
- z_{ω} : how far the actual angular rate $\dot{\vartheta}$ is from the desired rate α_1 computed in Step 2. If $z_{\omega} = 0$, the rocket is rotating at exactly the rate needed to drive z_{ϑ} to zero.
- z_{δ} : how far the actual nozzle angle δ is from the desired nozzle angle α_2 computed in Step 3. If $z_{\delta} = 0$, the nozzle is positioned exactly where needed to drive z_{ω} to zero.

Each error is only nonzero because the previous step's virtual control has not been fully achieved yet — this chain structure is the essence of backstepping.

Each virtual control (ϑ^* , α_1 , α_2) is chosen to make the corresponding new term in $\dot{V}$ negative-definite, while passing a residual cross-coupling to the next step.

6 Step 1 — Virtual Control 1: Desired Pitch ϑ^*

Compute $\dot{V}_{\text{pos}}$:

$$\dot{V}_{\text{pos}} = k_{px}e_x\dot{x} + \dot{x}\ddot{x} + k_{py}e_y\dot{y} + \dot{y}\ddot{y}$$

Choose the desired accelerations:

$$\begin{aligned} A_x &= -k_{px}e_x - k_{dx}\dot{x} \\ A_y &= -k_{py}e_y - k_{dy}\dot{y} \end{aligned} \quad (\text{O1})$$

and require $(F/m)\sin\vartheta^* = A_x$, $(F/m)\cos\vartheta^* = A_y + g$. This gives:

Remark 6.1 (Aerodynamics omitted from feedforward). The desired accelerations (O1) account only for thrust, not for aerodynamic forces X_b , Y_b . Including them would make the computation circular: computing ϑ^* requires knowing X_b , Y_b , which depend on the angle of attack $\alpha = \vartheta - \text{atan2}(\dot{x}, \dot{y})$, which in turn depends on ϑ^* . Instead, aerodynamic forces are treated as bounded disturbances — they appear as extra terms when (T1)–(T2) are substituted into $\dot{V}_{\text{pos}}$, and are absorbed into the ISS bound D of Section 10.5. This is justified because during the landing burn thrust dominates aerodynamics by roughly an order of magnitude, so the feedforward error is small and the feedback loop rejects the remainder.

$$\sigma = \text{clip}\left(\frac{m\sqrt{A_x^2 + (A_y + g)^2}}{F_{\text{max}}}, \sigma_{\text{min}}, 1\right) \quad (\text{O2})$$

$$\vartheta^* = \text{atan2}(A_x, A_y + g) \quad (\text{O3})$$

Hover check: $e_x = e_y = \dot{x} = \dot{y} = 0 \Rightarrow A_x = 0, A_y = 0, \sigma = mg/F_{\max}, \vartheta^* = 0$. ✓

Now substitute (T1)–(T2) and (O1) into $\dot{V}_{\text{pos}}$, applying A5: $\sin \vartheta \approx \sin \vartheta^* + \cos \vartheta^* \cdot z_\vartheta$, $\cos \vartheta \approx \cos \vartheta^* - \sin \vartheta^* \cdot z_\vartheta$. Using $(F/m) \sin \vartheta^* = A_x$ and $(F/m) \cos \vartheta^* = A_y + g$.

Derivation of the x -channel term. Substitute $\ddot{x}$ from (T1):

$$\ddot{x} + k_{px}e_x\dot{x} = \dot{x} \left(\frac{F \sin \vartheta + F\delta \cos \vartheta + X_b \sin \vartheta + Y_b \cos \vartheta}{m} + k_{px}e_x \right)$$

Apply A5 to expand $F \sin \vartheta/m$ around ϑ^* :

$$\frac{F \sin \vartheta}{m} \approx \frac{F \sin \vartheta^*}{m} + \frac{F \cos \vartheta^*}{m} z_\vartheta = A_x + \frac{F \cos \vartheta^*}{m} z_\vartheta$$

Collecting terms:

$$\ddot{x} + k_{px}e_x\dot{x} = \dot{x}(A_x + k_{px}e_x) + \dot{x} \frac{F}{m} \cos \vartheta^* \cdot z_\vartheta + \delta \frac{F}{m} \cos \vartheta \cdot \dot{x} + \frac{\dot{x}(X_b \sin \vartheta + Y_b \cos \vartheta)}{m}$$

With $A_x = -k_{px}e_x - k_{dx}\dot{x}$ from (O1), the first group gives $-k_{dx}\dot{x}^2$. The same expansion applied to the y -channel yields $-k_{dy}\dot{y}^2$ plus analogous residual terms.

The full aerodynamic contribution to $\dot{V}_{\text{pos}}$ from both channels is:

$$\frac{\dot{x}(X_b \sin \vartheta + Y_b \cos \vartheta) + \dot{y}(X_b \cos \vartheta - Y_b \sin \vartheta)}{m} = \frac{X_b(\dot{x} \sin \vartheta + \dot{y} \cos \vartheta) + Y_b(\dot{x} \cos \vartheta - \dot{y} \sin \vartheta)}{m}$$

The first group $\dot{x} \sin \vartheta + \dot{y} \cos \vartheta$ is the axial velocity V_a (velocity component along the body axis), so $X_b V_a/m = -C_x q_\infty S_m V_a/m$, which is non-positive in forward flight ($V_a \geq 0$). The second group $\dot{x} \cos \vartheta - \dot{y} \sin \vartheta$ is the lateral velocity V_n , and $Y_b V_n/m = C_{y\alpha} \alpha q_\infty S_m V_n/m$. This is not sign-definite in general; it is absorbed into the ISS bound D of Section 10.5 as an additional bounded term $\bar{d}_Y$.

Collecting the sign-definite terms:

$$\dot{V}_{\text{pos}} \leq -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 + z_\vartheta \underbrace{\frac{F}{m}(\dot{x} \cos \vartheta^* - \dot{y} \sin \vartheta^*)}_{\triangleq P} + \delta \underbrace{\frac{F}{m}(\dot{x} \cos \vartheta - \dot{y} \sin \vartheta)}_{\triangleq Q} + \bar{d}_Y \quad (\text{DV1})$$

where $\bar{d}_Y \geq 0$ is the bounded normal-force contribution. Note that P uses ϑ^* (from the A5 linearisation) while Q retains ϑ — Q is not linearised because it multiplies δ , a small quantity by A4, and is carried as a residual to be cancelled in Step 4.

The term $P = (F/m)(\dot{x} \cos \vartheta^* - \dot{y} \sin \vartheta^*)$ is the **position-attitude coupling** arising from the tilt-velocity interaction. The term $Q = (F/m)(\dot{x} \cos \vartheta - \dot{y} \sin \vartheta)$ is the **nozzle-position coupling** arising from retaining δ in (T1)–(T2).

7 Step 2 — Virtual Control 2: Desired Angular Rate α_1

Augment the Lyapunov function: $V_2 = V_{\text{pos}} + \frac{1}{2}z_\vartheta^2$.

Compute z_ϑ : $z_\vartheta = \dot{\vartheta} - \dot{\vartheta}^*$.

From (DV1) we already have $\dot{V}_{\text{pos}} \leq -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 + z_\vartheta P + \delta Q + \bar{d}_Y$. Adding the new $z_\vartheta \dot{z}_\vartheta$ term and grouping the z_ϑ contributions, substituting $\dot{\vartheta} = \alpha_1 + z_\omega$:

$$z_\vartheta(P + \dot{z}_\vartheta) = z_\vartheta(P + \dot{\vartheta} - \dot{\vartheta}^*) = z_\vartheta(P + \alpha_1 + z_\omega - \dot{\vartheta}^*)$$

Design step. Require $P + \alpha_1 - \dot{\vartheta}^* = -k_\vartheta z_\vartheta$:

$$\boxed{\alpha_1 = \dot{\vartheta}^* - k_\vartheta z_\vartheta - \frac{F}{m}(\dot{x} \cos \vartheta^* - \dot{y} \sin \vartheta^*)} \quad (\text{VC1})$$

Then $z_\vartheta(P + \alpha_1 + z_\omega - \dot{\vartheta}^*) = -k_\vartheta z_\vartheta^2 + z_\vartheta z_\omega$.

Result after Step 2:

$$\dot{V}_2 = -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 - k_\vartheta z_\vartheta^2 + z_\vartheta z_\omega + \delta \cdot Q + \bar{d}_Y \quad (\text{DV2})$$

The correction $-P = -(F/m)(\dot{x} \cos \vartheta^* - \dot{y} \sin \vartheta^*)$ in α_1 accounts for the fact that tilting the rocket changes the velocity components, which in turn affects the position error. Without this term, the $z_\vartheta z_\omega$ cross-coupling in $\dot{V}_2$ would not be cancelled and the Lyapunov inequality would have an uncompensated residual.

8 Step 3 — Virtual Control 3: Desired Nozzle Angle α_2

Augment: $V_3 = V_2 + \frac{1}{2}z_\omega^2$.

Compute $\dot{z}_\omega$: $\dot{z}_\omega = \ddot{\vartheta} - \dot{\alpha}_1 = g_2 \delta + f_2 - \dot{\alpha}_1$.

Time derivative:

$$\begin{aligned} \dot{V}_3 &= \dot{V}_2 + z_\omega \dot{z}_\omega \\ &= -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 - k_\vartheta z_\vartheta^2 + z_\vartheta z_\omega + z_\omega(g_2 \delta + f_2 - \dot{\alpha}_1) + \delta \cdot Q + \bar{d}_Y \end{aligned}$$

Grouping all δ -dependent terms, with $\Gamma \triangleq g_2 z_\omega + Q$:

$$\dot{V}_3 = -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 - k_\vartheta z_\vartheta^2 + z_\omega(z_\vartheta + f_2 - \dot{\alpha}_1) + \delta \cdot \Gamma + \bar{d}_Y \quad (\text{DV3})$$

Design step. Write $\delta = \alpha_2 + z_\delta$, so $\delta \cdot \Gamma = \alpha_2 \cdot \Gamma + z_\delta \cdot \Gamma$. Choose α_2 so that the z_ω bracket gives $-k_\omega z_\omega^2$:

$$z_\omega(z_\vartheta + f_2 - \dot{\alpha}_1 + g_2 \alpha_2) = -k_\omega z_\omega^2$$

$$\boxed{\alpha_2 = \frac{1}{g_2}(-k_\omega z_\omega - z_\vartheta - f_2 + \dot{\alpha}_1)} \quad (\text{VC2})$$

This requires $g_2 \neq 0$ — guaranteed by Assumption A6.

Result after Step 3:

$$\dot{V}_3 = -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 - k_\vartheta z_\vartheta^2 - k_\omega z_\omega^2 + z_\delta \cdot \Gamma + \alpha_2 \cdot Q + \bar{d}_Y \quad (\text{DV3}')$$

Two residual terms are passed to Step 4:

- $z_\delta \cdot \Gamma = z_\delta(g_2 z_\omega + Q)$: coupling between nozzle-angle error and the rest of the system — cancelled in Step 4.
- $\alpha_2 \cdot Q$: coupling between the virtual nozzle reference and the nozzle-position cross-term. At equilibrium ($\dot{x} = \dot{y} = 0$), $Q = 0$, so this term vanishes. It is bounded during the transient.

9 Step 4 — Real Control: Nozzle Command δ_{cmd}

Augment:

$$V = V_3 + \frac{1}{2}z_\delta^2 = \frac{1}{2}k_{px}e_x^2 + \frac{1}{2}\dot{x}^2 + \frac{1}{2}k_{py}e_y^2 + \frac{1}{2}\dot{y}^2 + \frac{1}{2}z_\vartheta^2 + \frac{1}{2}z_\omega^2 + \frac{1}{2}z_\delta^2 \quad (\text{V4})$$

Compute $\dot{z}_\delta$:

$$\dot{z}_\delta = \dot{\delta} - \dot{\alpha}_2 = \frac{-\delta + \delta_{\text{cmd}}}{\tau_\delta} - \dot{\alpha}_2 \quad (\text{D4})$$

Time derivative:

$$\begin{aligned} \dot{V} &= \dot{V}_3 + z_\delta \dot{z}_\delta \\ &= -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 - k_\vartheta z_\vartheta^2 - k_\omega z_\omega^2 + \alpha_2 \cdot Q + z_\delta \left(\Gamma + \frac{-\delta + \delta_{\text{cmd}}}{\tau_\delta} - \dot{\alpha}_2 \right) + \bar{d}_Y \end{aligned}$$

Design step. Choose δ_{cmd} to make the bracket multiplying z_δ equal to $-k_\delta z_\delta$:

$$\Gamma + \frac{-\delta + \delta_{\text{cmd}}}{\tau_\delta} - \dot{\alpha}_2 = -k_\delta z_\delta$$

Solving for δ_{cmd} :

$$\boxed{\delta_{\text{cmd}} = \delta + \tau_\delta(\dot{\alpha}_2 - g_2 z_\omega - Q - k_\delta z_\delta)} \quad (\text{CL})$$

Verification of cancellation. Substituting (CL) into the z_δ bracket:

$$\Gamma + \frac{-\delta + \delta_{\text{cmd}}}{\tau_\delta} - \dot{\alpha}_2 = (g_2 z_\omega + Q) + (\dot{\alpha}_2 - g_2 z_\omega - Q - k_\delta z_\delta) - \dot{\alpha}_2 = -k_\delta z_\delta \quad \checkmark$$

Apply saturation: $\delta_{\text{cmd}} = \text{clip}(\delta_{\text{cmd}}, -\delta_{\text{max,cmd}}, \delta_{\text{max,cmd}})$.

The term $-Q = -(F/m)(\dot{x} \cos \vartheta - \dot{y} \sin \vartheta)$ in (CL) feeds back the translational velocities into the nozzle command to compensate the position-nozzle coupling $\delta \cdot Q$ that appears in (DV1).

$\dot{V}$ after substituting (CL):

$$\dot{V} = -k_{dx}\dot{x}^2 - k_{dy}\dot{y}^2 - k_\vartheta z_\vartheta^2 - k_\omega z_\omega^2 - k_\delta z_\delta^2 + \alpha_2 \cdot Q + \bar{d}_Y \quad (\text{ISS1})$$

10 Stabilization Proof

10.1 Handling the $\alpha_2 \cdot Q$ Residual

Before proceeding to the proof, we characterise the single term in (ISS1) that does not have a definite sign.

The coupling $Q = (F/m)(\dot{x} \cos \vartheta - \dot{y} \sin \vartheta)$ satisfies $|Q| \leq (F/m)\sqrt{\dot{x}^2 + \dot{y}^2}$. Applying Young's inequality with parameter $\epsilon > 0$:

$$|\alpha_2 \cdot Q| \leq \frac{\epsilon}{2}\alpha_2^2 + \frac{1}{2\epsilon}Q^2 \leq \frac{\epsilon}{2}\alpha_2^2 + \frac{(F/m)^2}{2\epsilon}(\dot{x}^2 + \dot{y}^2)$$

Choosing ϵ such that:

$$\epsilon \geq \frac{(F/m)^2}{2 \min(k_{dx}, k_{dy})} \quad (\text{C-eps})$$

the Q^2 residual is absorbed by the velocity damping terms, and (ISS1) becomes:

$$\dot{V} \leq -\left(k_{dx} - \frac{(F/m)^2}{2\epsilon}\right)\dot{x}^2 - \left(k_{dy} - \frac{(F/m)^2}{2\epsilon}\right)\dot{y}^2 - k_{\vartheta}z_{\vartheta}^2 - k_{\omega}z_{\omega}^2 - k_{\delta}z_{\delta}^2 + \frac{\epsilon}{2}\alpha_2^2 \quad (\text{ISS2})$$

Define $k'_{dx} = k_{dx} - (F/m)^2/(2\epsilon) > 0$ and $k'_{dy} = k_{dy} - (F/m)^2/(2\epsilon) > 0$ (both positive by (C-eps)). The remaining term $\frac{\epsilon}{2}\alpha_2^2$ is positive but vanishes at equilibrium since $\alpha_2 \rightarrow 0$ when $z_{\omega}, z_{\vartheta}, f_2 \rightarrow 0$ in (VC2). Its treatment is in Section 10.2.

10.2 Boundedness of All Signals

Note on the $\frac{\epsilon}{2}\alpha_2^2$ term. From (VC2), $\alpha_2 = \frac{1}{g_2}(-k_{\omega}z_{\omega} - z_{\vartheta} - f_2 + \dot{\alpha}_1)$, so α_2 is a function of $z_{\omega}, z_{\vartheta}, f_2$, and $\dot{\alpha}_1$, all of which are components of the state vector covered by V . Taking the absolute value and using $g_2 \geq g_{2,\min} > 0$ (Assumption A6):

$$|\alpha_2| \leq \frac{1}{g_{2,\min}}(k_{\omega}|z_{\omega}| + |z_{\vartheta}| + |f_2| + |\dot{\alpha}_1|)$$

Since $V \geq \frac{1}{2}z_{\vartheta}^2$ and $V \geq \frac{1}{2}z_{\omega}^2$:

$$|z_{\vartheta}| \leq \sqrt{2V} \quad (7)$$

$$|z_{\omega}| \leq \sqrt{2V} \quad (8)$$

Boundedness of $|f_2|$. The aerodynamic moment $f_2 = C_{m\alpha} \alpha q_{\infty} S_m l / J$ depends on airspeed $V = \sqrt{\dot{x}^2 + \dot{y}^2}$ (bounded by physical limits of the scenario) and angle of attack $\alpha = \vartheta - \text{atan2}(\dot{x}, \dot{y})$. The pitch ϑ is bounded via $\vartheta = \vartheta^* + z_{\vartheta}$, where $\vartheta^* = \text{atan2}(A_x, A_y + g)$ depends only on the bounded position errors $e_x, \dot{x}, e_y, \dot{y}$ (covered by $V_{\text{pos}} \leq V$). Therefore:

$$|f_2| \leq \frac{C_{m\alpha} |\alpha|_{\max} q_{\infty, \max} S_m l}{J} \triangleq \bar{f}_2 < \infty$$

Boundedness of $|\dot{\alpha}_1|$. Differentiating (VC1): $\dot{\alpha}_1 = \ddot{\vartheta}^* - k_{\vartheta}z_{\vartheta} - \dot{P}$. Each term depends on positions, velocities, and their derivatives — all bounded by V and the physical equations of motion. Therefore $|\dot{\alpha}_1| \leq \bar{\alpha}_1 < \infty$.

Combining all four bounds, there exists a constant C_{α_2} depending only on $V(0)$ and the gains such that:

$$\frac{\epsilon}{2}\alpha_2^2 \leq \frac{\epsilon}{2}C_{\alpha_2}^2 \triangleq \mu$$

Substituting into (ISS2):

$$\dot{V} \leq -k'_{dx}\dot{x}^2 - k'_{dy}\dot{y}^2 - k_{\vartheta}z_{\vartheta}^2 - k_{\omega}z_{\omega}^2 - k_{\delta}z_{\delta}^2 + \mu \quad (\text{ISS3})$$

Theorem 10.1 (UUB). *Under the control law (CL), virtual controls (VC1)–(VC2), and Assumptions A1–A7, if gains satisfy (C-eps) strictly, then all signals $(e_x, \dot{x}, e_y, \dot{y}, z_{\vartheta}, z_{\omega}, z_{\delta})$ are **uniformly ultimately bounded**: they enter and remain in the compact set $\Omega = \{V \leq \mu/c\}$ in finite time, where:*

$$c = \min(k'_{dx}, k'_{dy}, k_{\vartheta}, k_{\omega}, k_{\delta})$$

The transient bound is $V(t) \leq \max(V(0), \mu/c) \triangleq V_{\max}$. Individual bounds:

$$|e_x(t)| \leq \sqrt{\frac{2V_{\max}}{k_{px}}} \triangleq B_{e_x} \quad (9)$$

$$|\dot{x}(t)| \leq \sqrt{2V_{\max}} \triangleq B_{\dot{x}} \quad (10)$$

and analogously for $e_y, \dot{y}, z_{\vartheta}, z_{\omega}, z_{\delta}$. All bounds depend only on $V(0)$, μ , and the gains.

Consequence. Since $(z_\vartheta, z_\omega, z_\delta)$ are bounded and the virtual controls (VC1)–(VC2) are continuous functions of bounded signals, $\vartheta, \dot{\vartheta}, \delta$ and α_2 are all bounded. In particular, the singularity $g_2 = 0$ is never reached under A6.

Remark 10.1 (Two-phase stability picture). The overall stability result has two phases:

- **Phase 1 — transient** ($\alpha_2 \neq 0, \mu > 0$): While the system is away from equilibrium, $\alpha_2 \neq 0$ and the residual $\mu = \frac{\epsilon}{2}C_{\alpha_2}^2 > 0$ from the $\alpha_2 \cdot Q$ cross-term is nonzero. Additionally, clipping of $\alpha_1, \alpha_2, \delta_{\text{cmd}}$ may be active (e.g. during the first 1–2 s with a 30° initial tilt), contributing the extra disturbance D' of Section 10.5. In this phase only UUB holds: all errors are bounded and remain in the ball $\Omega = \{V \leq D'/c\}$. The system does not diverge, but exact convergence to zero is not guaranteed.
- **Phase 2 — steady descent** ($\alpha_2 \rightarrow 0, \mu \rightarrow 0$): As the system approaches equilibrium, $\alpha_2 \rightarrow 0$ so $\mu \rightarrow 0$, and clipping becomes inactive so $D' \rightarrow 0$. The stronger results of Sections 10.3 and 10.4 apply: Barbalat gives $\dot{x}, \dot{y}, z_\vartheta, z_\omega, z_\delta \rightarrow 0$, and LaSalle gives $e_x, e_y \rightarrow 0$.

In short: the controller is practically stable throughout, and asymptotically convergent once α_2 is small and clipping is inactive.

10.3 Convergence of Velocities and Attitude Errors — Barbalat’s Lemma

Remark 10.2 (Scope). The Barbalat argument below is an idealised analysis that assumes $\mu = 0$ from the start — i.e., the $\alpha_2 \cdot Q$ cross-term is neglected. This is a standard simplification used to establish the convergence structure of the controller. In reality $\mu = \frac{\epsilon}{2}C_{\alpha_2}^2 > 0$ whenever $\alpha_2 \neq 0$, so the true result is UUB (Section 10.2), not exact convergence to zero. The Barbalat argument is nonetheless useful because it shows what the system would do in the absence of the residual, and because μ is small when α_2 is small (i.e., when the nozzle is near its equilibrium position).

Theorem 10.2 (Idealised, $\mu = 0$). *If the $\alpha_2 \cdot Q$ residual is neglected, then $\dot{x}(t) \rightarrow 0, \dot{y}(t) \rightarrow 0, z_\vartheta(t) \rightarrow 0, z_\omega(t) \rightarrow 0, z_\delta(t) \rightarrow 0$ as $t \rightarrow \infty$.*

Proof. We show the argument for $\dot{x}$; the others follow identically.

Step 1 — Integrability. With $\mu = 0$, (ISS3) gives $\dot{V} \leq -k'_{dx} \dot{x}^2$. Integrating from 0 to T :

$$\int_0^T k'_{dx} \dot{x}^2(\tau) d\tau \leq V(0) - V(T) \leq V(0) < \infty$$

Taking $T \rightarrow \infty$: $\dot{x}^2 \in L^2([0, \infty))$.

Step 2 — Uniform continuity. From Section 10.2, $|\dot{x}| \leq \sqrt{2V_{\text{max}}}$. From (T1), $\ddot{x}$ is a continuous function of bounded signals, so $\frac{d}{dt}(\dot{x}^2) = 2\dot{x}\ddot{x}$ is bounded, meaning $\dot{x}^2$ is uniformly continuous.

Step 3 — Barbalat’s Lemma. A non-negative uniformly continuous function with finite L^2 integral converges to zero:

$$\dot{x}^2(t) \rightarrow 0 \implies \dot{x}(t) \rightarrow 0 \quad \text{as } t \rightarrow \infty. \quad \square$$

□

The same argument applies to $\dot{y}, z_\vartheta, z_\omega$, and z_δ using the corresponding negative-definite terms in (ISS3).

10.4 Convergence of Position Errors — LaSalle’s Invariance Principle

Theorem 10.3 (Idealised, same scope as Section 10.3). *If $\mu = 0$, then $e_x(t) \rightarrow 0$ and $e_y(t) \rightarrow 0$ as $t \rightarrow \infty$. Under the true $\mu > 0$, position errors are bounded by $|e_x| \leq B_{e_x}$ from Section 10.2; exact convergence to zero is not claimed.*

Proof. Applicability of LaSalle. The closed-loop system is **autonomous** — the right-hand side $f(s)$ depends only on the state s , not explicitly on t (the controller is a pure state-feedback law with no external time-varying signals). LaSalle’s invariance principle (Khalil, Theorem 4.4) therefore applies directly. The set $\{\dot{V} = 0\}$ contains states where $\dot{x} = \dot{y} = z_\vartheta = z_\omega = z_\delta = 0$ but e_x, e_y may be non-zero, so we must identify the largest invariant subset of $\{\dot{V} = 0\}$. All trajectories remain in the compact sublevel set $\{V \leq V_{\max}\}$ (Section 10.2), satisfying the compactness requirement of the theorem.

Define the limiting set:

$$\mathcal{S} = \{(e_x, \dot{x}, e_y, \dot{y}, z_\vartheta, z_\omega, z_\delta) : \dot{x} = \dot{y} = z_\vartheta = z_\omega = z_\delta = 0\}$$

In $\mathcal{S}$ the following hold simultaneously:

- $\dot{x} = \dot{y} = 0 \Rightarrow \ddot{x} = \ddot{y} = 0$; also $Q = 0$.
- $z_\vartheta = 0 \Rightarrow \vartheta = \vartheta^* = \text{atan2}(A_x, A_y + g)$.
- $z_\omega = 0 \Rightarrow \dot{\vartheta} = \alpha_1$; with $\dot{x} = \dot{y} = 0$ and $z_\vartheta = 0$: $\alpha_1 = \dot{\vartheta}^* - 0 - 0 = \dot{\vartheta}^*$, so $\dot{\vartheta} = \dot{\vartheta}^*$.
- $z_\delta = 0 \Rightarrow \delta = \alpha_2$.

Substituting $\dot{x} = \dot{y} = 0$ into (T1) (aerodynamic terms vanish since $V = 0$):

$$m\ddot{x} = F \sin \vartheta^*$$

By construction (O3), $(F/m) \sin \vartheta^* = A_x$, so $F \sin \vartheta^* = mA_x$, giving $m\ddot{x} = mA_x$.

But $\ddot{x} = 0$ in $\mathcal{S}$, so:

$$0 = A_x = -k_{px}e_x - k_{dx}\underbrace{\dot{x}}_{=0} = -k_{px}e_x \implies e_x = 0$$

The same argument applied to (T2) gives $e_y = 0$.

Therefore the largest invariant subset of $\mathcal{S}$ is $\{e_x = e_y = 0\}$, and $(e_x(t), e_y(t)) \rightarrow 0$. $\square$ $\square$

10.5 Stability Under Saturation — ISS Analysis

Section 10.2 (UUB) holds regardless of clipping. The idealised convergence proofs in Sections 10.3–10.4 additionally assume that α_1, α_2 , and δ_{cmd} are delivered exactly. In the real implementation all three are clipped:

$$\alpha_{1,\text{sat}} = \text{clip}(\alpha_{1,\text{raw}}, -\alpha_{1,\text{max}}, \alpha_{1,\text{max}}) \tag{11}$$

$$\alpha_{2,\text{sat}} = \text{clip}(\alpha_{2,\text{raw}}, -\delta_{\text{max}}, \delta_{\text{max}}) \tag{12}$$

$$\delta_{\text{cmd,sat}} = \text{clip}(\delta_{\text{cmd}}, -\delta_{\text{max,cmd}}, \delta_{\text{max,cmd}}) \tag{13}$$

This section proves that the closed-loop system remains **input-to-state stable (ISS)** with respect to the saturation errors, giving a rigorous global result without assuming saturation never occurs.

10.5.1 Saturation Errors as Bounded Disturbances

Each saturation is modelled as an additive bounded disturbance. All three disturbances are uniformly bounded regardless of state.

α_1 **saturation:**

$$d_{\vartheta} = \alpha_{1,\text{sat}} - \alpha_{1,\text{raw}}, \quad |d_{\vartheta}| \leq 2\alpha_{1,\text{max}} \triangleq \bar{d}_{\vartheta} \quad (14)$$

This enters the z_{ω} dynamics as $-d_{\vartheta}$ (since $z_{\omega} = \dot{\vartheta} - \alpha_1$ and α_1 is clipped).

α_2 **saturation:**

$$d_{\omega} = \alpha_{2,\text{sat}} - \alpha_{2,\text{raw}}, \quad |d_{\omega}| \leq 2\delta_{\text{max}} \quad (15)$$

This propagates through the z_{ω} dynamics with gain g_2 , giving bounded disturbance $|g_2 d_{\omega}| \leq g_{2,\text{max}} \cdot 2\delta_{\text{max}} \triangleq \bar{d}_{\omega}$.

δ_{cmd} **saturation:**

$$d_{\delta} = \delta_{\text{cmd},\text{sat}} - \delta_{\text{cmd}}, \quad |d_{\delta}| \leq 2\delta_{\text{max},\text{cmd}} \quad (16)$$

This enters the z_{δ} dynamics divided by τ_{δ} , giving bounded disturbance $|d_{\delta}/\tau_{\delta}| \leq 2\delta_{\text{max},\text{cmd}}/\tau_{\delta} \triangleq \bar{d}_{\delta}$.

10.5.2 Perturbed Error Dynamics

Substituting $\alpha_{2,\text{sat}} = \alpha_{2,\text{raw}} + d_{\omega}$ and using the design choice (VC2):

$$\dot{z}_{\omega} = -k_{\omega} z_{\omega} - z_{\vartheta} + g_2 z_{\delta} + g_2 d_{\omega} - d_{\vartheta}$$

Substituting $\delta_{\text{cmd},\text{sat}} = \delta_{\text{cmd}} + d_{\delta}$ and using (CL):

$$\dot{z}_{\delta} = -k_{\delta} z_{\delta} - g_2 z_{\omega} - Q + \frac{d_{\delta}}{\tau_{\delta}}$$

10.5.3 Lyapunov Derivative Under Disturbances

The z_{ω} and z_{δ} contributions to $\dot{V}$ are:

$$z_{\omega} \dot{z}_{\omega} = -k_{\omega} z_{\omega}^2 - z_{\vartheta} z_{\omega} + g_2 z_{\omega} z_{\delta} + g_2 d_{\omega} z_{\omega} - d_{\vartheta} z_{\omega}$$

$$z_{\delta} \dot{z}_{\delta} = -k_{\delta} z_{\delta}^2 - g_2 z_{\omega} z_{\delta} - Q z_{\delta} + \frac{d_{\delta}}{\tau_{\delta}} z_{\delta}$$

The $g_2 z_{\omega} z_{\delta}$ terms cancel exactly. The cross-terms are bounded via Young's inequality:

- $|g_2 d_{\omega} z_{\omega}| \leq \frac{k_{\omega}}{8} z_{\omega}^2 + \frac{2g_2^2 \bar{d}_{\omega}^2}{k_{\omega}} \quad (\text{with } \lambda_1 = k_{\omega}/4)$
- $\left| \frac{d_{\delta}}{\tau_{\delta}} z_{\delta} \right| \leq \frac{k_{\delta}}{16} z_{\delta}^2 + \frac{4\bar{d}_{\delta}^2}{k_{\delta}\tau_{\delta}^2} \quad (\text{with } \lambda_2 = k_{\delta}/8)$
- $|d_{\vartheta} z_{\omega}| \leq \frac{k_{\omega}}{8} z_{\omega}^2 + \frac{2\bar{d}_{\vartheta}^2}{k_{\omega}} \quad (\text{with } \lambda_5 = k_{\omega}/4)$

These reduce the z_{ω}^2 coefficient by $k_{\omega}/8 + k_{\omega}/8 = k_{\omega}/4$ total and the z_{δ}^2 coefficient by $k_{\delta}/16$, keeping both positive. The $z_{\vartheta} z_{\omega}$ cross-term (from (DV2)) is bounded with $\lambda_4 = k_{\vartheta}/2$:

$$|z_{\vartheta} z_{\omega}| \leq \frac{k_{\vartheta}}{4} z_{\vartheta}^2 + \frac{1}{k_{\vartheta}} z_{\omega}^2$$

provided $k_\omega > 1/k_\vartheta + 1$. The Qz_δ term is absorbed into the velocity terms via (C-eps) extended to cover $\lambda_3 = k_\delta/4$.

Collecting all terms:

$$\dot{V} \leq -k''_{dx} \dot{x}^2 - k''_{dy} \dot{y}^2 - \frac{k_\vartheta}{2} z_\vartheta^2 - \frac{k_\omega}{4} z_\omega^2 - \frac{k_\delta}{8} z_\delta^2 + D' \quad (\text{ISS4})$$

where the total disturbance bound is:

$$D' = \mu + \frac{2g_{2,\max}^2 \bar{d}_\omega^2}{k_\omega} + \frac{2\bar{d}_\vartheta^2}{k_\omega} + \frac{4\bar{d}_\delta^2}{k_\delta \tau_\delta^2} \quad (17)$$

10.5.4 UUB Conclusion

Theorem 10.4 (UUB under saturation). *Under Assumptions A1–A7, gain condition (C-eps), and $k_\omega > 1/k_\vartheta + 1$, the closed-loop system with saturated controls is **uniformly ultimately bounded**: all states enter and remain in $\mathcal{B} = \{V \leq D/c\}$, where:*

$$c = \min\left(k''_{dx}, k''_{dy}, \frac{k_\vartheta}{2}, \frac{k_\omega}{4}, \frac{k_\delta}{8}\right) \quad (18)$$

$$D = D' \quad (19)$$

The residual bounds on attitude errors are:

$$|z_\omega|_\infty \leq \sqrt{\frac{2D}{c k_\omega}} \quad (20)$$

$$|z_\delta|_\infty \leq \sqrt{\frac{2D}{c k_\delta}} \quad (21)$$

10.5.5 Quantitative Bound for the Project Parameters

With parameters from `configs/backstepping.yaml`:

- $g_{2,\max} = F_{\max} l_{cp}/J = 191763 \times 10.5/318196.65 \approx 6.33 \text{ rad/s}^2/\text{rad}$
- $\alpha_{1,\max} = 1.5 \text{ rad/s}$, so $\bar{d}_\vartheta = 3.0 \text{ rad/s}$
- $\delta_{\max} = 15^\circ = 0.262 \text{ rad}$, so $\bar{d}_\omega = 2 \times 6.33 \times 0.262 \approx 3.31 \text{ rad/s}^2$
- $\delta_{\max,\text{cmd}} = 0.262 \text{ rad}$, $\tau_\delta = 0.05 \text{ s}$, so $\bar{d}_\delta \approx 10.47 \text{ rad/s}$

With $k_\omega = 5$, $k_\delta = 15$, $\tau_\delta = 0.05 \text{ s}$, the three contributions to D' are:

$$\frac{2g_{2,\max}^2 \bar{d}_\omega^2}{k_\omega} \approx 175.8 \quad (22)$$

$$\frac{2\bar{d}_\vartheta^2}{k_\omega} \approx 3.6 \quad (23)$$

$$\frac{4\bar{d}_\delta^2}{k_\delta \tau_\delta^2} \approx 11697 \quad (24)$$

The δ_{cmd} clipping term dominates — it is amplified by $1/\tau_\delta^2 = 400$. Total $D' \approx \mu + 11876$.

With $c = \min(k_\vartheta/2, k_\omega/4, k_\delta/8) = \min(3.0, 1.25, 1.875) = 1.25$:

$$|z_\omega|_\infty \leq \sqrt{\frac{2 \times 11876}{1.25 \times 5}} \approx 61.6 \text{ rad/s} \approx 3532^\circ/\text{s}$$

This absurdly large bound is a direct consequence of the $1/\tau_\delta^2$ amplification — the ISS framework assumes δ_{cmd} clipping is permanently active, which never happens in practice. In the simulation, δ_{cmd} clipping clears within 1–2 s; after that $d_\delta = 0$ exactly, $D' \rightarrow \mu$, and the system converges asymptotically. The bound is technically correct but extremely conservative.

This large bound assumes all three clipping saturations are **permanently active simultaneously**. In the simulation, α_1 clipping clears within 0.1 s and α_2 clipping within 1–2 s; after that $d_\vartheta = d_\omega = d_\delta = 0$ exactly. As the system then converges, $\alpha_2 \rightarrow 0$ so $\mu \rightarrow 0$, and the results of Sections 10.3–10.4 apply, giving true asymptotic convergence to zero.

11 Full Algorithm

State: $s = [x, y, \dot{x}, \dot{y}, \vartheta, \dot{\vartheta}, \delta]^T$. Filter state: α_2^f (command filter — introduced in Block E to avoid differentiating α_2 analytically). Derivative filter states: $\dot{\vartheta}_{\text{filt}}, \dot{\alpha}_{1,\text{filt}}$ (ODE states, see Section 13).

Block A — Step 1: Virtual control 1 (ϑ^*, σ)

```

1. e_x = x - x_d,    e_y = y
2. A_x = -k_px * e_x - k_dx * xdot
3. A_y = -k_py * e_y - k_dy * ydot
4. a_vert = max(A_y + g, |A_x| / tan(phi_lim), a_min)    [safety clamp]
5. A_y = a_vert - g
6. A_x = clip(A_x, -a_vert*tan(phi_lim), a_vert*tan(phi_lim))
7. T_des = sqrt(A_x^2 + a_vert^2)
8. sigma = clip(m * T_des / F_max, sigma_min, 1)
9. F = sigma * F_max
10. theta* = atan2(A_x, a_vert)
11. theta*_dot ~= filtered analytically (see Section 13)

```

Remark 11.1 (Safety clamping, steps 4–6). The theoretical design requires only $a_{\min} = 0$ for the Lyapunov proof; the implementation enforces $a_{\min} = 0.5 \text{ m/s}^2$ and the tilt limit $\phi_{\text{lim}} = 30^\circ$ to keep the rocket from commanding full free-fall or excessive tilt. The clamp on A_x ensures $|\vartheta^*| \leq \phi_{\text{lim}}$ exactly. These modifications introduce a bounded error in (O1) handled by Section 10.5.

Block B — Aerodynamics

```

12. V_spd = sqrt(xdot^2 + ydot^2)
13. alpha_aoa = (V_spd < V_min) ? theta : theta - atan2(xdot, ydot)
14. q_inf = 0.5 * rho * V_spd^2
15. f2 = C_ma * alpha_aoa * q_inf * S_m * l / J
16. g2 = F * l_cp / J

```

Block C — Step 2: Virtual control 2 (α_1)

```

17. z_theta = theta - theta*
18. alpha1 = theta*_dot - k_phi * z_theta
19. z_omega = theta_dot - alpha1
20. alpha1_dot ~= (alpha1(t) - alpha1(t-dt)) / dt

```

Remark 11.2 (P-coupling omission). The theoretical formula (VC1) includes $-(F/m)(\dot{x} \cos \vartheta^* - \dot{y} \sin \vartheta^*)$. In practice this reaches 60+ rad/s at high entry velocities and permanently saturates α_1 . The simplified form $\alpha_1 = \dot{\vartheta}^* - k_\vartheta z_\vartheta$ is used instead. The omitted P term is a bounded disturbance covered by Section 10.5.

Remark 11.3 (Filtered derivatives). The implementation propagates $\dot{\vartheta}^*$ and $\dot{\alpha}_1$ as ODE states driven by first-order filters toward their instantaneous analytical values (bandwidth $N = 50$ rad/s). The lag error is bounded by $O(1/N)$ and constitutes a bounded disturbance within the ISS framework.

Remark 11.4 (α_1 saturation). The implementation clips α_1 to $\pm\alpha_{1,\max}$ (default 1.5 rad/s). This introduces a bounded disturbance in the z_ω dynamics analogous to the α_2 saturation analysed in Section 10.5, with $\bar{d}_{\alpha_1} = 2\alpha_{1,\max}$.

Block D — Step 3: Virtual control 3 (α_2)

```
21. alpha2_raw = (1/g2) * (-k_omega * z_omega - z_theta - f2 +
    alpha1_dot)
22. alpha2_sat = clip(alpha2_raw, -delta_max, delta_max)
```

Block E — Command filter

```
23. alpha2f_dot = (alpha2_sat - alpha2f) / tau_f
24. alpha2f     += alpha2f_dot * dt      [or integrate inside ODE]
25. z_delta     = delta - alpha2f
```

Remark 11.5 (Command filtering). The Lyapunov derivation defines $z_\delta = \delta - \alpha_2$ and uses $\dot{\alpha}_2$ in (CL). In the implementation, α_2 is replaced by its filtered version α_2^f , so $z_\delta = \delta - \alpha_2^f$ and $\dot{\alpha}_2^f$ enters (CL). The filter error $e_f = \alpha_2^f - \alpha_2$ satisfies $\dot{e}_f = -e_f/\tau_f + (\alpha_{2,\text{sat}} - \alpha_2)/\tau_f$, which is bounded. Command filtering (Farrell et al., 2005) avoids differentiating α_2 analytically; the resulting filter error introduces an additional bounded disturbance of order $O(\tau_f)$ absorbed by Section 10.5.

Block F — Step 4: Real control (δ_{cmd})

```
26. delta_cmd = delta + tau_delta * (alpha2f_dot - g2*z_omega - k_delta*
    z_delta)
27. delta_cmd = clip(delta_cmd, -delta_max_cmd, delta_max_cmd)
```

Remark 11.6 (Q-coupling omission). The theoretical formula (CL) includes $-Q = -(F/m)(\dot{x} \cos \vartheta - \dot{y} \sin \vartheta)$. This term reaches 50+ rad/s at high entry velocities and, multiplied by τ_δ , produces nozzle commands far exceeding $\delta_{\max,\text{cmd}}$. The term is omitted in the implementation; the bounded error is covered by Section 10.5.

Output: $[\sigma, \delta_{\text{cmd}}]$

12 Gain Conditions and Tuning Guidelines

Necessary conditions

$$k_{px}, k_{dx}, k_{py}, k_{dy}, k_\vartheta, k_\omega, k_\delta > 0, \quad \sigma_{\min} > 0, \quad \tau_f > 0$$

Critical damping of position loop

$$k_{dx} \geq 2\sqrt{k_{px}} \quad (25)$$

$$k_{dy} \geq 2\sqrt{k_{py}} \quad (26)$$

Attitude loop bandwidth

No formal time-scale separation is required, but faster attitude convergence reduces the time during which the $\alpha_2 \cdot Q$ residual is active. A practical guideline:

$$\sqrt{k_\vartheta g_{2,\text{hov}}} \geq 3\sqrt{k_{px}} \quad (27)$$

$$g_{2,\text{hov}} = \frac{\sigma_{\text{hov}} F_{\max} l_{cp}}{J} \quad (28)$$

Cross-term damping condition

From the $z_\vartheta z_\omega$ bounding step in Section 10.5:

$$k_\omega > \frac{1}{k_\vartheta} + 1$$

For the project gains $k_\vartheta = 6$: $k_\omega > 1/6 + 1 \approx 1.17$. Current value $k_\omega = 5$ satisfies this with large margin.

Residual $\alpha_2 \cdot Q$ condition (C-eps)

From Section 10.1, the Young's inequality bound on $|\alpha_2 \cdot Q|$ introduces the term $(F/m)^2(\dot{x}^2 + \dot{y}^2)/(2\epsilon)$, which must be absorbed by the velocity damping. This requires:

$$\frac{(F/m)^2}{2\epsilon} \leq \min(k_{dx}, k_{dy}) \iff \epsilon \geq \frac{(F/m)^2}{2 \min(k_{dx}, k_{dy})}$$

For the project parameters, $(F/m)_{\max} = F_{\max}/m = 191763/13000 \approx 14.75 \text{ m/s}^2$, so $\epsilon \geq 14.75^2/(2 \times 5.0) \approx 21.7$. Since ϵ is a free parameter in Young's inequality (not a gain to tune), this condition is always satisfiable.

Command filter time constant

$$\frac{\tau_\delta}{10} \lesssim \tau_f \lesssim \frac{\tau_\delta}{5}$$

13 Implementation Notes

Ten-state ODE

```
state = [x, y, vx, vy, phi, omega, delta, alpha2f, ts_dot, a1_dot]
```

The last two states are first-order filtered derivatives: `ts_dot` tracks ϑ^* and `a1_dot` tracks $\dot{\alpha}_1$, driven analytically inside the ODE.

Filtered derivatives

ϑ^* and $\dot{\alpha}_1$ are needed by (VC2) and (CL). The implementation propagates them as ODE states driven by first-order filters toward their analytical instantaneous values:

```
ts_dot_instant = ((A_y+g)*dAx_dt - A_x*dAy_dt) / max(A_x**2+(A_y+g)**2,
a_vert**2)
ts_dot_instant = clip(ts_dot_instant, -alpha1_max, alpha1_max)
a1_dot_instant = ts_dot_instant - k_phi * (omega - ts_dot)

d/dt(ts_dot) = N_deriv * (ts_dot_instant - ts_dot)    # N_deriv = 50 rad/s
d/dt(a1_dot) = N_deriv * (a1_dot_instant - a1_dot)
```

This avoids differentiating ϑ^* numerically (which is noisy), at the cost of a lag error bounded by $O(1/N_{\text{deriv}})$.

Angle wrapping

```
z_phi = (phi - theta_star + np.pi) % (2 * np.pi) - np.pi
```


Zero-airspeed guard

```
if V_spd < V_min:    # V_min ~= 0.1 m/s
    alpha_aoa, q_inf = phi, 0.0
else:
    alpha_aoa = phi - np.arctan2(vx, vy)
    q_inf = 0.5 * rho * V_spd**2
```

Minimum throttle guard

```
sigma = max(sigma, sigma_min)
F = sigma * F_max
g2 = F * l_cp / J    # guaranteed > 0
```

Python variable mapping

Symbol	Python variable
ϑ	phi
$\dot{\vartheta}$	omega
ϑ^*	theta_star
δ	delta
δ_{cmd}	delta_cmd
α_2^f	alpha2f
$z_\vartheta, z_\omega, z_\delta$	z_phi, z_omega, z_delta
α_1, α_2	alpha1, alpha2
P, Q	P_coupling, Q_coupling
f_2, g_2	f2, g2
σ	sigma
$k_\vartheta, k_\omega, k_\delta$	k_phi, k_omega, k_delta
$k_{px}, k_{dx}, k_{py}, k_{dy}$	k_px, k_dx, k_py, k_dy
τ_δ, τ_f	tau_delta, tau_filter

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